# Additional engineering drawings from the web

10 drawing views/sheets across 5 source documents | Supplement to the existing collection

The shaft and connecting-rod papers provide drawings and numerical FEM results.

The wheel and hoists provide detailed drawings, but no matching FEM results.

Contents

1 shaft drawing with static and rotating-load FEM references

2 connecting-rod drawings with material-specific FEM stress references

5 wheel sheets: assembly plus four dimensioned components

2 manufacturer hoist sheets with assembly envelope dimensions

These are supplementary candidates, not 10 new fully reproducible CAD/FEM pairs.

No native CAD or raw FEM file was verified for these exact drawings.

Original drawings are reproduced as source excerpts, with clickable attribution.

# Centrifugal-pump stepped shaft

Dimensioned drawing | Published FEM values available | Native CAD/raw fields unverified

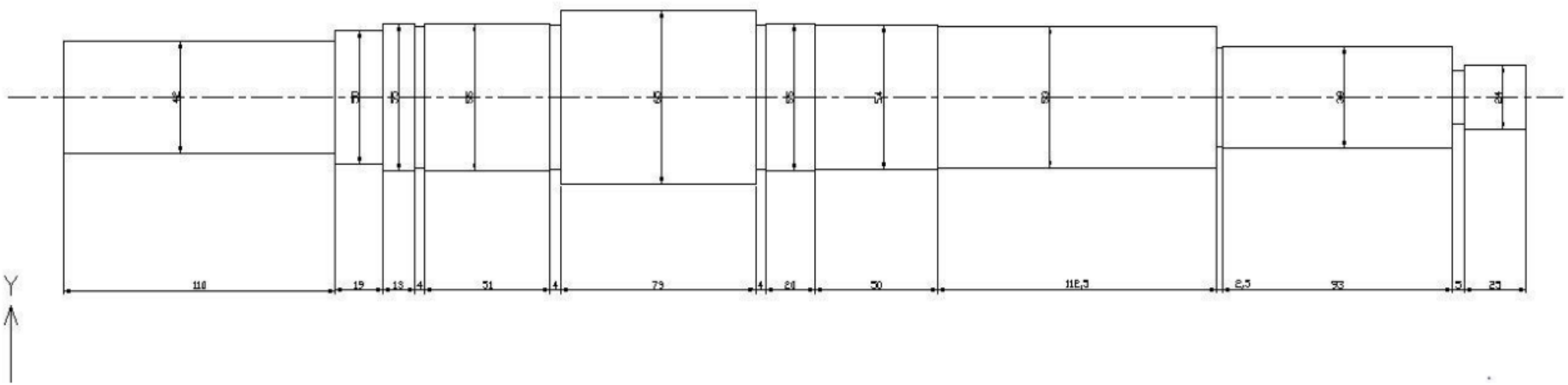



Fig.1 cad drawing of shaft.

Dimensioned shaft drawing, overall length 593 mm; source also describes slots/keyways.

Static FEM: downward displacement -4.838 micrometres; maximum stress 2.23 MPa.

Rotating-load FEM: maximum displacement 0.783 mm; von Mises stress 238 MPa.

The source treats rotating loads with a static solve; not a transient simulation.

Native CAD / solver deck / raw result fields: not verified.

Solver naming is inconsistent: RADIOSS in methods; Nastran in conclusion.

Source: https://www.iosrjournals.org/iosr-jmce/papers/vol7-issue3/F0733742.pdf

# Pump shaft: published FEM evidence

Original source figure excerpts | Paper pages 3 and 5

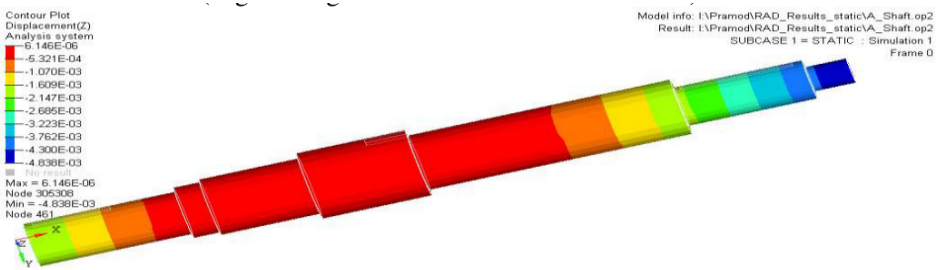



Fig.7.displacement

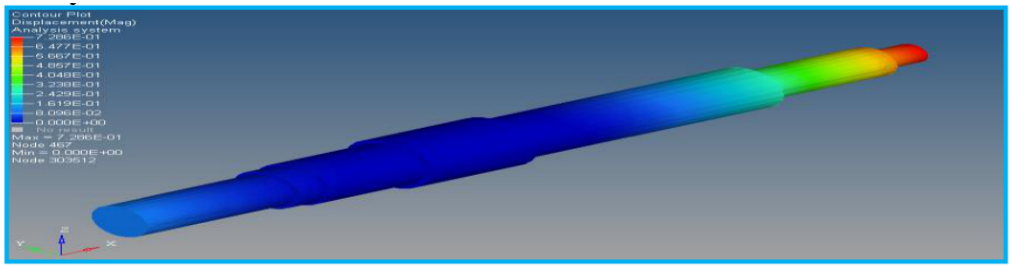



Fig.12. dynamic deflection

Static: -4.838 um downward displacement; maximum stress 2.23 MPa.

Rotating-load case: maximum displacement 0.783 mm; von Mises stress 238 MPa.

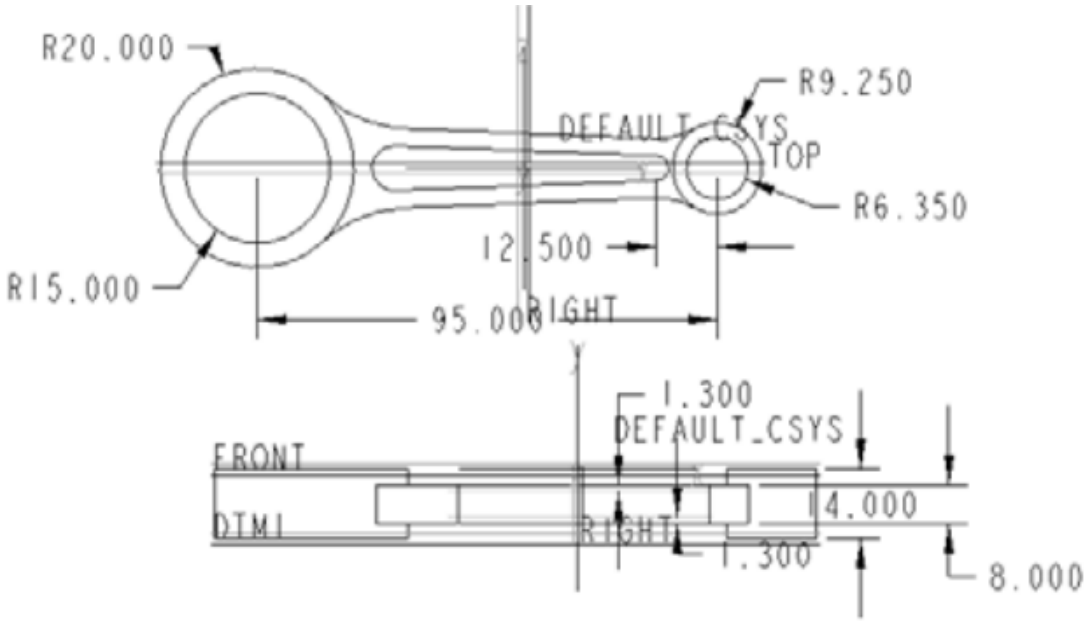
These values are transcribed from the source, not recomputed.

Loads, assumptions and solver inconsistencies remain in the source paper.

Source: https://www.iosrjournals.org/iosr-jmce/papers/vol7-issue3/F0733742.pdf

# Connecting rod: Original

Dimensioned drawing | Published material-specific FEM results | Paper page 3




Two dimensioned orthographic drawings, with material tables and FEM comparisons.

Compressive bearing load: 4319 N; source states crank fixed in each analysis.

E-glass/epoxy stress: original 96.135 MPa; modified I 80.995 MPa.

Kevlar49/epoxy stress: original 155.31 MPa; modified I 121.95 MPa.

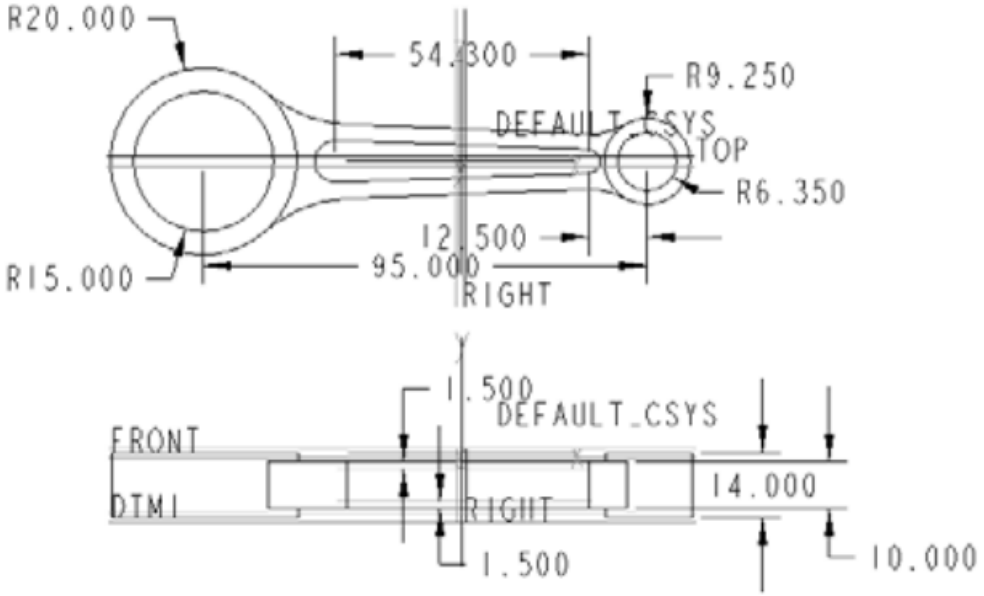
Values refer to Table 4, not the separate validation model in Table 1.

Native CAD / solver deck / raw result fields: not verified.

Source: https://globaljournals.org/GJRE_Volume14/6-Design-Optimization-and-Comparative-Stress.pdf

# Connecting rod: Modified I

Dimensioned drawing | Published material-specific FEM results | Paper page 3




Two dimensioned orthographic drawings, with material tables and FEM comparisons.

Compressive bearing load: 4319 N; source states crank fixed in each analysis.

E-glass/epoxy stress: original 96.135 MPa; modified I 80.995 MPa.

Kevlar49/epoxy stress: original 155.31 MPa; modified I 121.95 MPa.

Values refer to Table 4, not the separate validation model in Table 1.

Native CAD / solver deck / raw result fields: not verified.

Source: https://globaljournals.org/GJRE_Volume14/6-Design-Optimization-and-Comparative-Stress.pdf

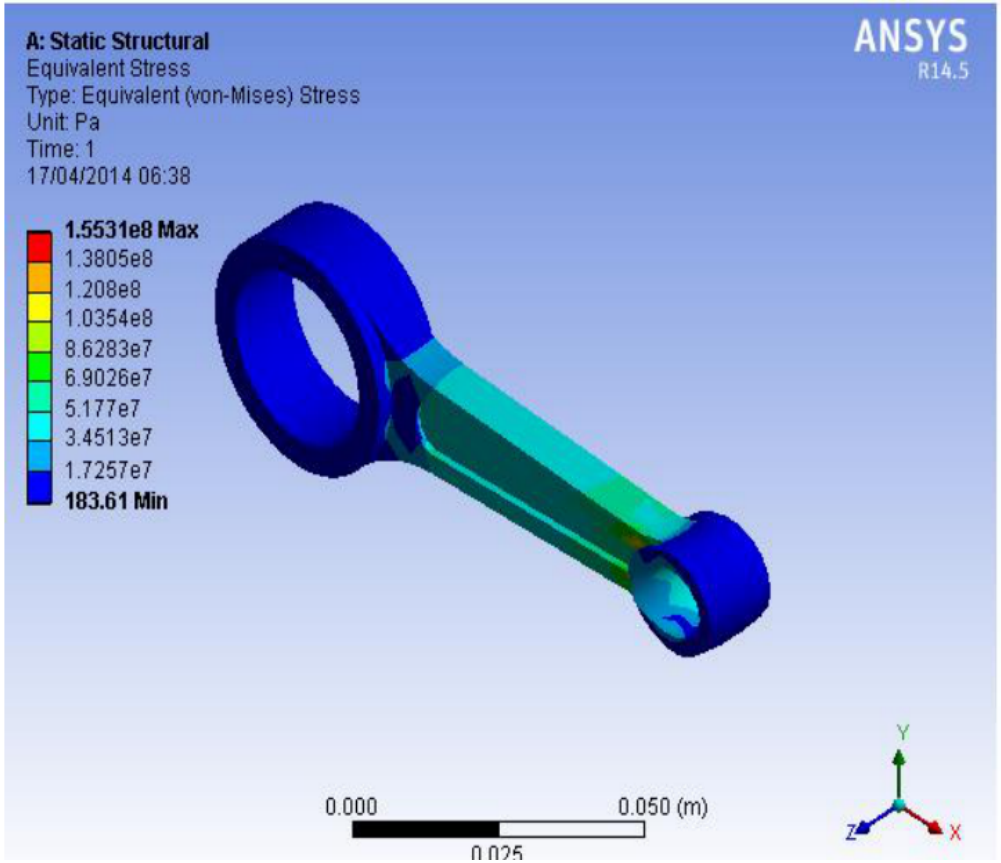
# Connecting rod: published FEM evidence

Original Table 4 and equivalent-stress contour excerpt

*Table 4 :* For Compressive bearing load 4319 N and Maximum equivalent von-mises stress

| Parameters | Eglass/ Epoxy (Mpa) | Kevlar49/ Epoxy (Mpa) |
|---|---|---|
| Existing Model | 96.135 | 155.31 |
| Modified Model I | 80.995 | 121.95 |
| Variation*(%) | 15.75 | 21.479 |
| Modified Model 2 | 84.567 | 127.03 |
| Variation*(%) | 12.033 | 18.20 |
| Modified Model3 | 87.05 | 125.31 |
| Variation*(%) | 9.45 | 19.31 |

**Variation (%) with existing model.*




Load: 4319 N. Match each reference value to its material and geometry variant.

Source: https://globaljournals.org/GJRE_Volume14/6-Design-Optimization-and-Comparative-Stress.pdf

Published source illustrations; no locally generated CAD or FEM.

# Assembly / BOM

Original drawing sheet | FEM results not provided by this source

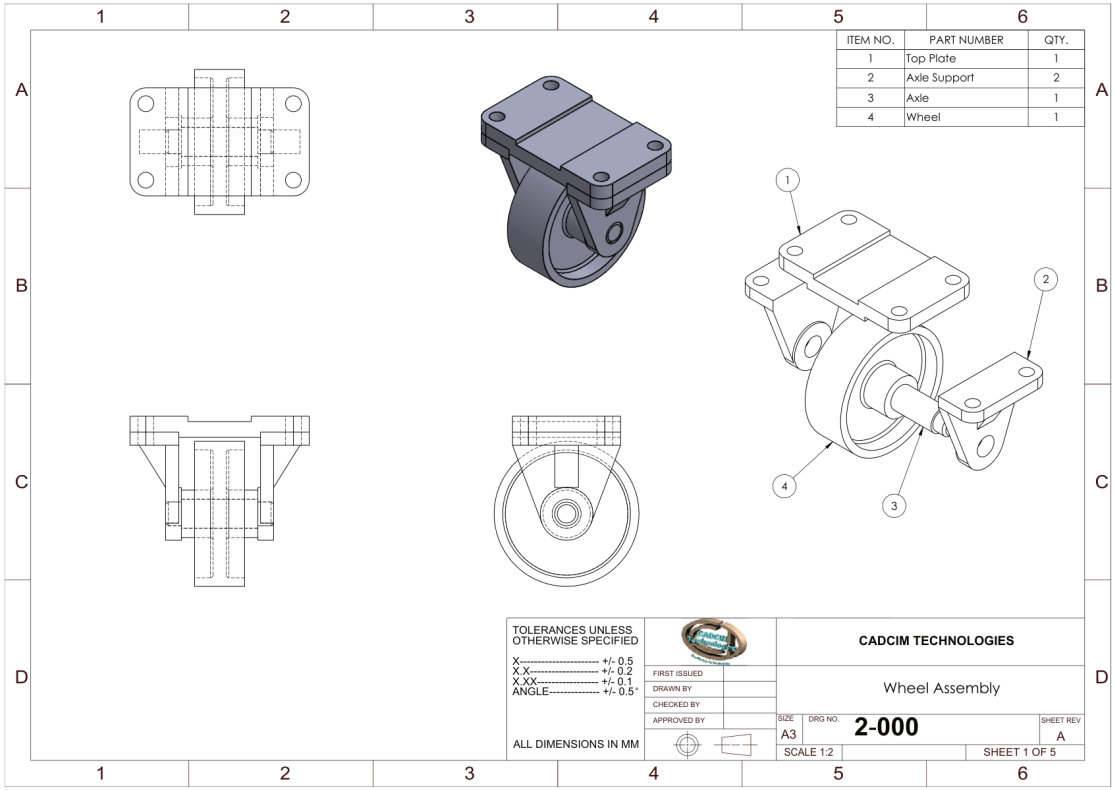



*Figure P2-1* *The Wheel assembly*

P2-2

CATIA V5-6R2020 for Designers

Source document page 2. Native CAD and matching FEM results for this sheet remain unverified.

Source: https://www.cadcim.com/978-1-64057-108-2/Projects/Project-2.pdf

Published source illustrations; no locally generated CAD or FEM.

# Axle support

Original drawing sheet | FEM results not provided by this source

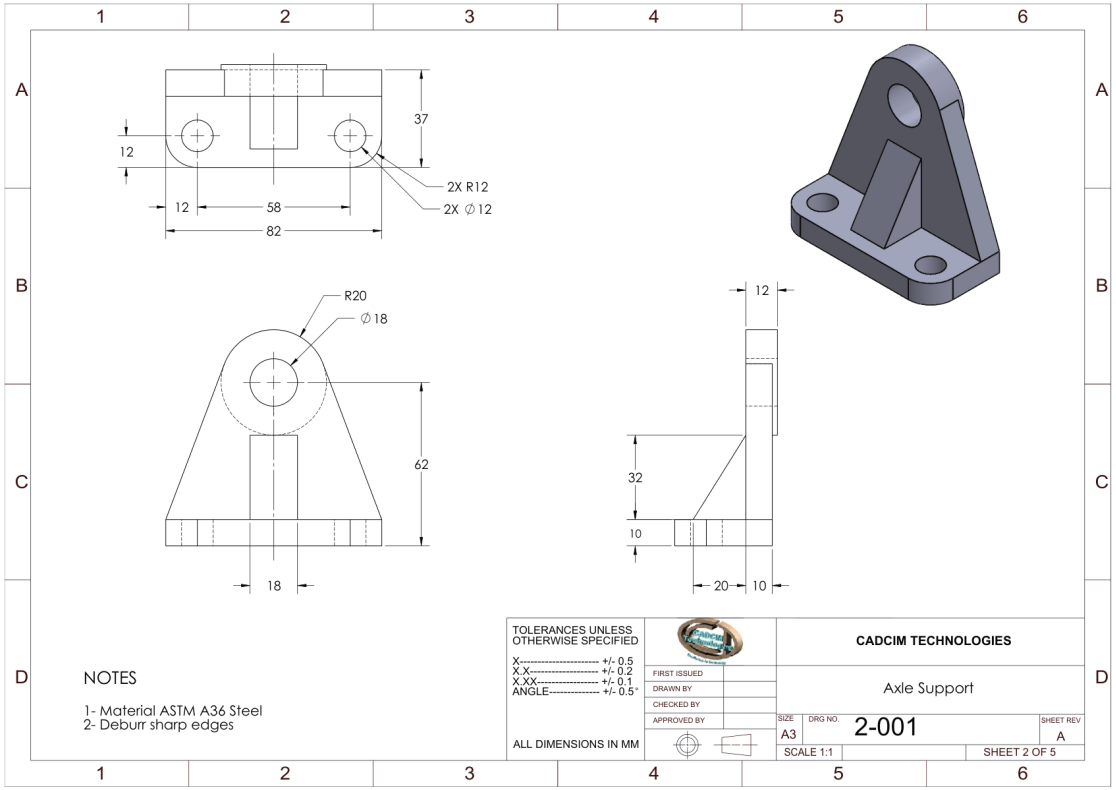



***Figure P2-2*** *Views and dimensions of Axle Support*

Source document page 3. Native CAD and matching FEM results for this sheet remain unverified.

Source: https://www.cadcim.com/978-1-64057-108-2/Projects/Project-2.pdf

Published source illustrations; no locally generated CAD or FEM.

# Axle

Original drawing sheet | FEM results not provided by this source

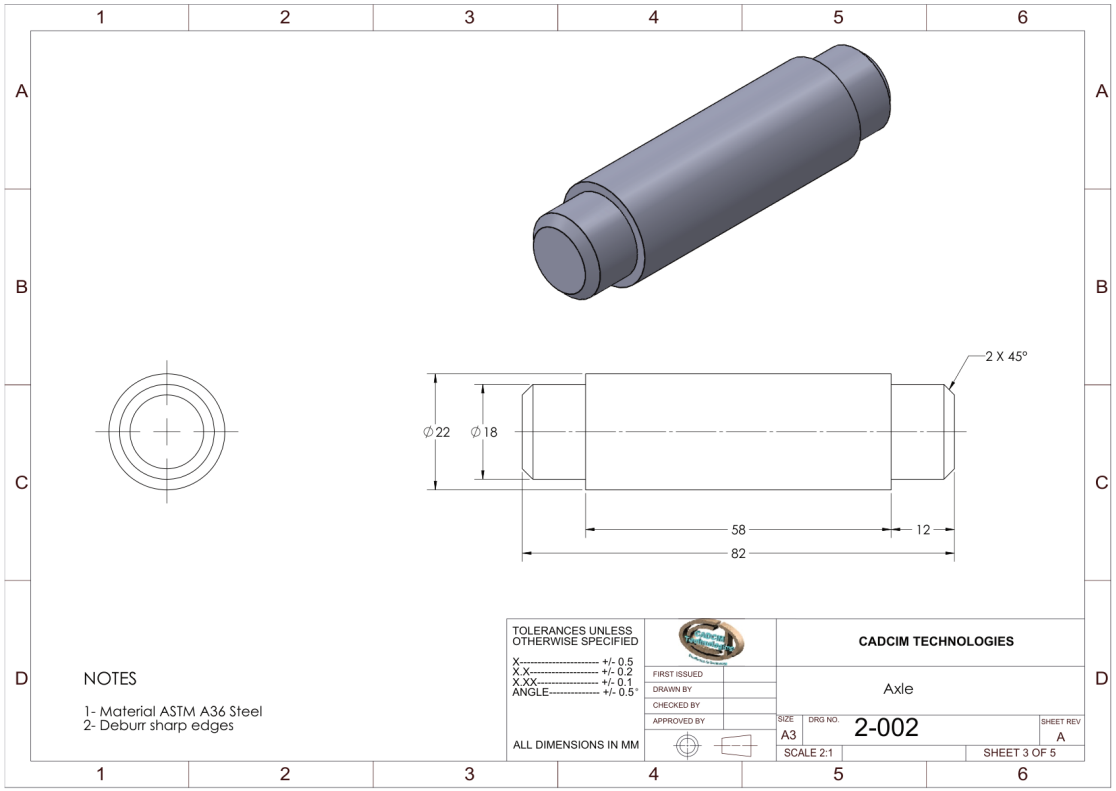



*Figure P2-3* *Views and dimensions of Axle*

P2-4

CATIA V5-6R2020 for Designers

Source document page 4. Native CAD and matching FEM results for this sheet remain unverified.

Source: https://www.cadcim.com/978-1-64057-108-2/Projects/Project-2.pdf

Published source illustrations; no locally generated CAD or FEM.

# Top plate

Original drawing sheet | FEM results not provided by this source

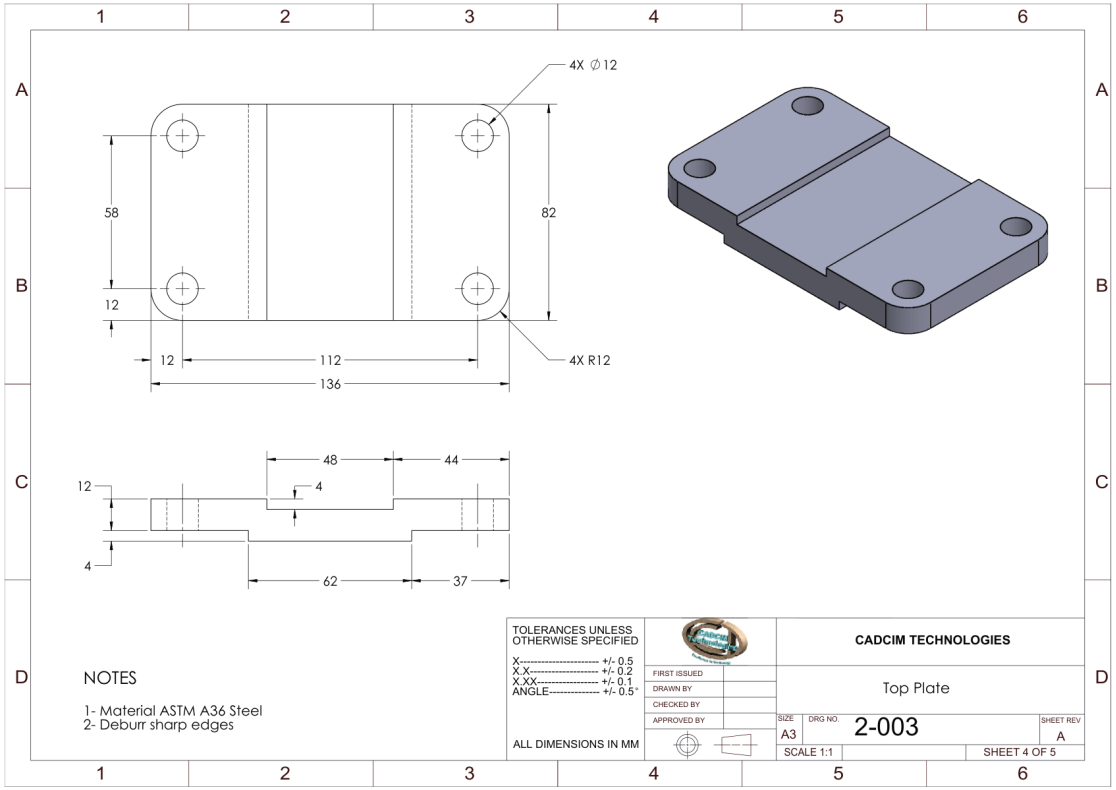



***Figure P2-4*** *Views and dimensions of Top Plate*

Project 2

P2-5

Source document page 5. Native CAD and matching FEM results for this sheet remain unverified.

Source: https://www.cadcim.com/978-1-64057-108-2/Projects/Project-2.pdf

Published source illustrations; no locally generated CAD or FEM.

# Wheel

Original drawing sheet | FEM results not provided by this source

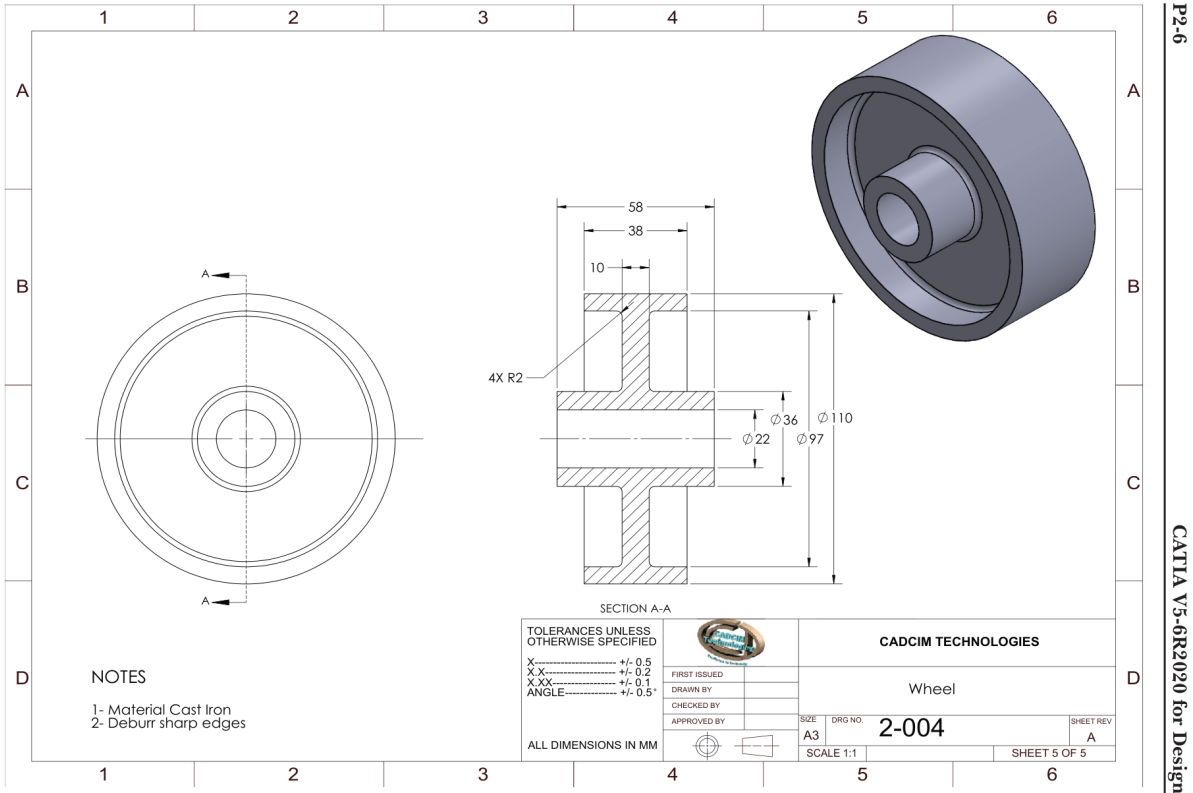



***Figure P2-5*** *Views and dimensions of Wheel*

Source document page 6. Native CAD and matching FEM results for this sheet remain unverified.

Source: https://www.cadcim.com/978-1-64057-108-2/Projects/Project-2.pdf

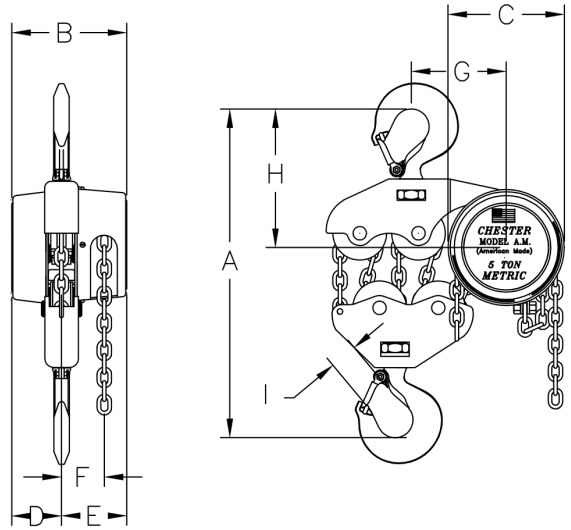
Published source illustrations; no locally generated CAD or FEM.

# Chester 5 metric ton hook hoist

Original drawing sheet | FEM results not provided by this source

NO. REQ.

**FOR MORE INFO CONTACT:**
**American Crane & Equipment Corp.**
**Authorized Distributor**
**Tel: 877-503-2972**
**Fax: 484-945-0430**
**Email: sales@americancrane.com**
**www.americancrane.com**




| A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|
| 20 1/4" | 6 1/2" | 6 1/4" | 2 13/16" | 3 11/16" | 2 1/2" | 4 3/8" | 7 7/8" | 1 3/8" |

| MK. | REVISIONS | CK–DATE |
|---|---|---|
| | | |

5 METRIC TON "AM" HOIST

CHESTER HOIST
LISBON, OHIO

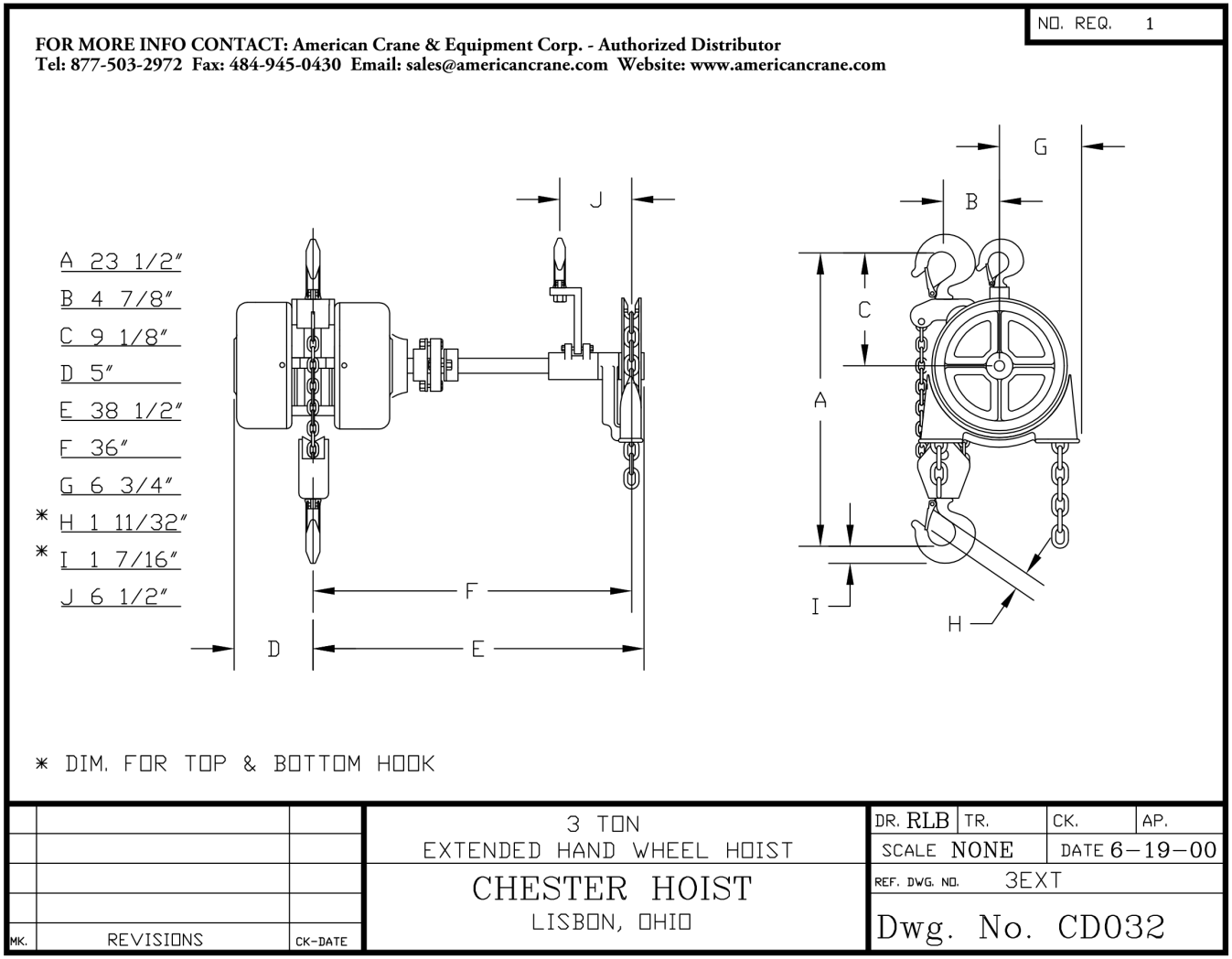
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Dwg. No. CD072

Source document page 1. Native CAD and matching FEM results for this sheet remain unverified.

Source: https://americancrane.com/wp-content/uploads/2024/06/AM_Hook_Type_150-5_DWG.pdf

# Chester 3 ton extended handwheel hoist

Original drawing sheet | FEM results not provided by this source




Source document page 1. Native CAD and matching FEM results for this sheet remain unverified.

Source: https://americancrane.com/wp-content/uploads/2024/06/EXTENDED_HANDWHEEL_134-3.pdf

Published source illustrations; no locally generated CAD or FEM.